

Coffee Shop Customer & Sales Analysis — Analytics Report

Business Analytics Report

Project: Coffee Shop Customer & Sales Analysis

Dataset: coffee_shop_transactions.csv

Analysis date: 2026-09-25

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Generated by the AI Data Scientist Platform

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1. Executive Summary

This report analyzes **coffee_shop_transactions.csv** (69,785 rows, 12 columns) covering 2025-01-01 to 2099-01-01 to answer the business question: *A synthetic transaction log of a coffee chain with five branches in Karachi, Lahore, and Islamabad, covering January 2025 to March 2026. It contains about 69,800 purchase lines with customer ID, date and time, branch, item, category, quantity, price, discount, payment method, and loyalty points. It includes refunds, walk-in customers without IDs, duplicates, missing values, prices stored as text, inconsistent branch names, and invalid dates.*

Key verified findings

- The original dataset contained 6,523 missing value(s) and 45 duplicate row(s), which have since been cleaned.
- A Random Forest model was trained to predict **Unit Price Pkr** (regression). On average, predictions are off by about 0.00 units (Mean Absolute Error).

Model reliability, in plain terms: On average, predictions are off by about 0.00 units (Mean Absolute Error). The model explains about 100.0% of the variation in the target ($R^2 = 1.000$).

Top recommendations

- 'Total Pkr' is the strongest single predictor of Unit Price Pkr in this model (relative importance 63.1%).
- 'Category: Bakery' is the second most influential predictor of Unit Price Pkr in this model (relative importance 13.4%).
- 'Category: Hot Coffee' is also a meaningful predictor of Unit Price Pkr in this model (relative importance 9.5%).

2. About the Dataset

coffee_shop_transactions.csv contains 69,785 rows and 12 columns. It spans the period 2025-01-01 to 2099-01-01 (based on Transaction Time).

Business question: A synthetic transaction log of a coffee chain with five branches in Karachi, Lahore, and Islamabad, covering January 2025 to March 2026. It contains about 69,800 purchase lines with customer ID, date and time, branch, item, category, quantity, price, discount, payment method, and loyalty points. It includes refunds, walk-in customers without IDs, duplicates, missing values, prices stored as text, inconsistent branch names, and invalid dates.

Column dictionary

Column	Type	Missing	Notes
Transaction Id	Identifier	0.0%	
Customer Id	Identifier	0.0%	most common: C6188
Transaction Time	Date/Time	0.0%	
Branch	Category	0.0%	most common: Karachi - DHA
Item	Category	0.0%	most common: Blueberry Muffin
Category	Category	0.0%	most common: Hot Coffee
Quantity	Numeric	0.06%	typical range 1.00–3.00
Unit Price Pkr	Numeric	0.0%	typical range 420.00–1,026.00
Discount Pkr	Numeric	0.0%	typical range 0.00–308.00
Payment Method	Category	0.0%	most common: Card
Loyalty Points Redeemed	Numeric	0.0%	typical range 0.00–299.00

Total Pkr	Numeric	0.06%	typical range 378.00–3,078.
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3. Data Quality & Cleaning

Original dataset quality (as uploaded)

Total rows	Missing cells	Duplicate rows	Issues detected
69,830	6,523	45	7

Duplicate Rows

Column	Finding	Affected
(whole dataset)	45 duplicate row(s) detected.	45

Missing Values

Column	Finding	Affected
customer_id	'customer_id' has 4429 missing value(s) (6.3%).	4429
payment_method	'payment_method' has 2094 missing value(s) (3.0%).	2094

Invalid Dates

Column	Finding	Affected
transaction_time	'transaction_time' has 4 value(s) that don't parse as a valid date.	4

Impossible Values

Column	Finding	Affected
quantity	'quantity' has 44 negative value(s) that are rare exceptions among otherwise non-negative values — likely a data-entry error, not real signal.	44
total_pkr	'total_pkr' has 44 negative value(s) that are rare exceptions among otherwise non-negative values — likely a data-entry error, not real signal.	44

Outliers

Column	Finding	Affected
total_pkr	'total_pkr' has 8270 statistical outlier(s) outside [87.5, 1467.5].	8270

Cleaning performed

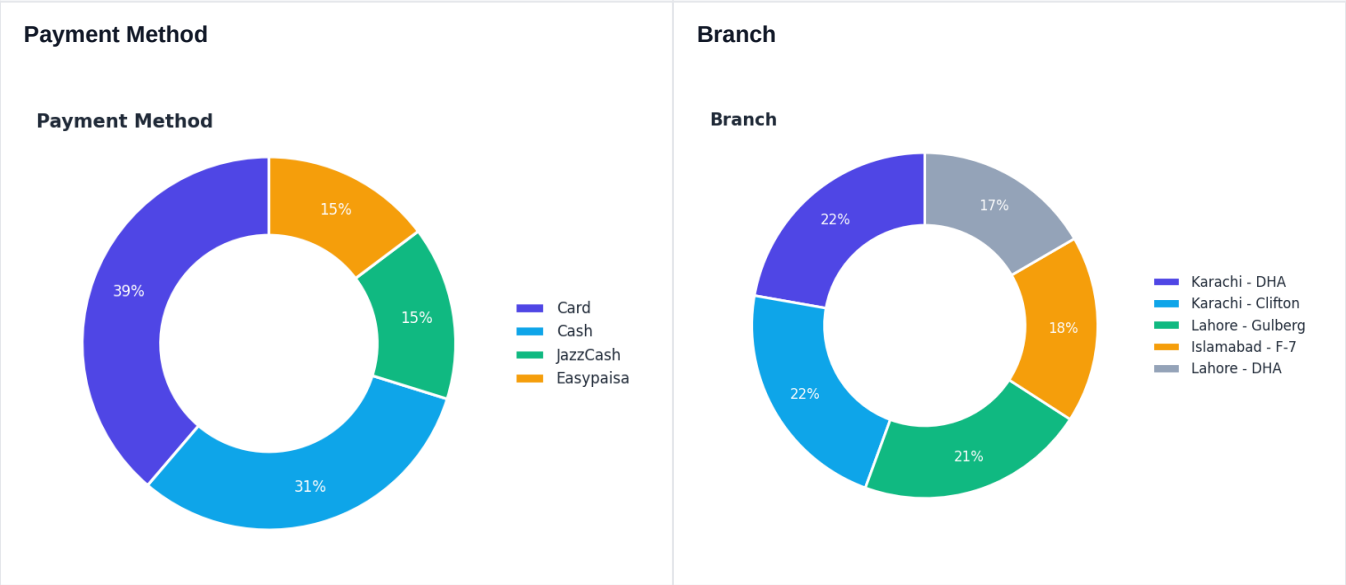
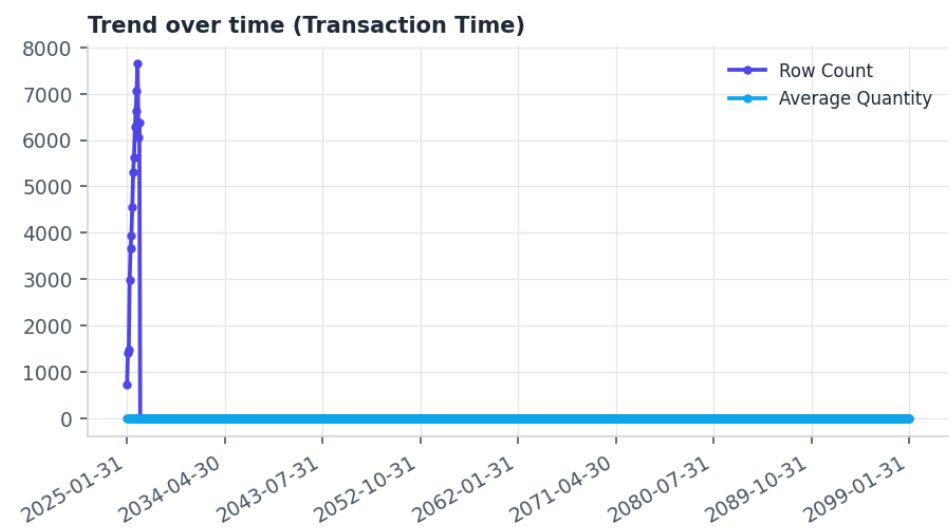
Problem	Before	Action	After	Reason
Duplicate rows	45 duplicate row(s)	Removed duplicate rows	0 duplicate rows	Duplicate records would otherwise double-count the same transaction/customer.
Missing Customer Id	4429 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Missing Payment Method	2094 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Impossible negative Quantity	44 impossible value(s)	Converted to missing, then filled	0 impossible values	A rare negative value in an otherwise non-negative column is almost always a data-entry error.
Impossible negative Total Pkr	44 impossible value(s)	Converted to missing, then filled	0 impossible values	A rare negative value in an otherwise non-negative column is almost always a data-entry error.
Text-formatted numbers in Unit Price Pkr	stored as text with units (e.g. '4.2 km')	Converted to a clean numeric value	numeric	100% of values matched a number-with-unit pattern.
Inconsistent capitalization in Branch	5 variant label(s) (e.g. 'rainy' vs 'Rainy')	Merged into a single canonical label	consistent labels	Values differing only by letter case would otherwise be treated as different categories.

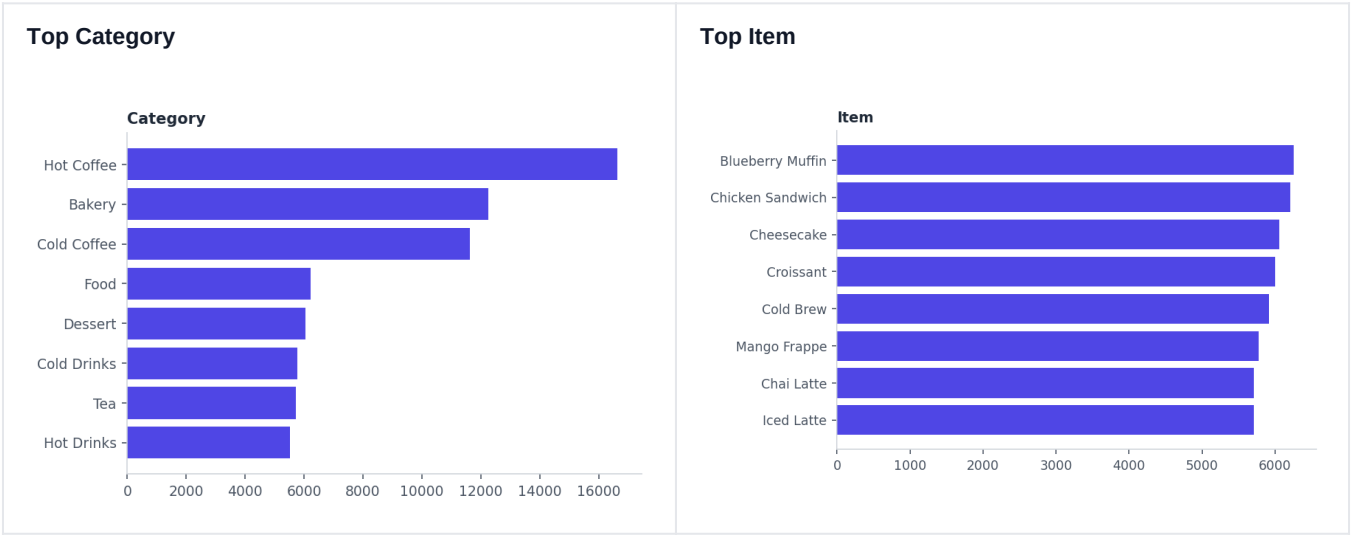
After cleaning: 88 missing cell(s), 0 duplicate row(s), 69,785 rows remain. The original file itself was never modified — cleaning produced a separate working copy.

4. Key EDA Findings

Total Rows	Average Unit Price Pkr	Total Quantity	Total Discount Pkr
69,785	716	87,068	679,961

Trend over time (Transaction Time)





Quantity breakdown

Average: 1.25 (range 1.0–3.0)

Value	Count	Percent
3	3473	5.0%
2	10381	14.9%
1	55887	80.1%

5. Modeling Approach

A **regression** model was trained to predict **Unit Price Pkr**, using 11 feature(s).

Features used

Transaction Time: Year, Transaction Time: Month, Transaction Time: Dayofweek, Branch, Item, Category, Quantity, Discount Pkr, Payment Method, Loyalty Points Redeemed, Total Pkr

Features excluded, and why

Column	Reason excluded
Transaction Id	Identifier column (e.g. an ID) — carries no predictive signal.
Customer Id	Identifier column (e.g. an ID) — carries no predictive signal.
Transaction Time	Raw date — replaced with derived year / month / day-of-week features instead.

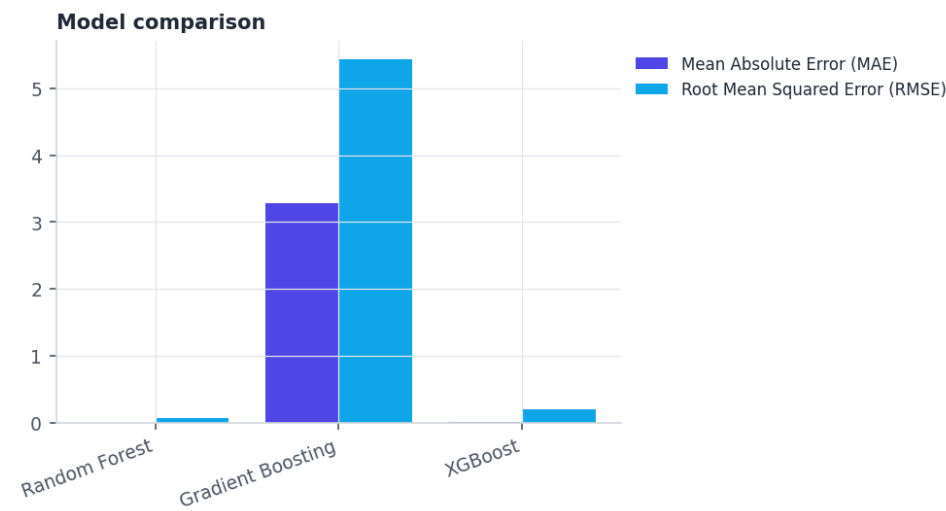
Technical details

Algorithm: Random Forest · Preprocessing: 7 numeric + 4 categorical feature(s), scaled/encoded · Train/test split: 20.0% held out · Cross-validation folds: 5 · Random seed: 42

6. Model Performance

Model comparison

Model	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	R ² (variance explained)
Random Forest	0.0007	0.0718	1.0000
Gradient Boosting	3.2778	5.4426	0.9991
XGBoost	0.0044	0.1995	1.0000

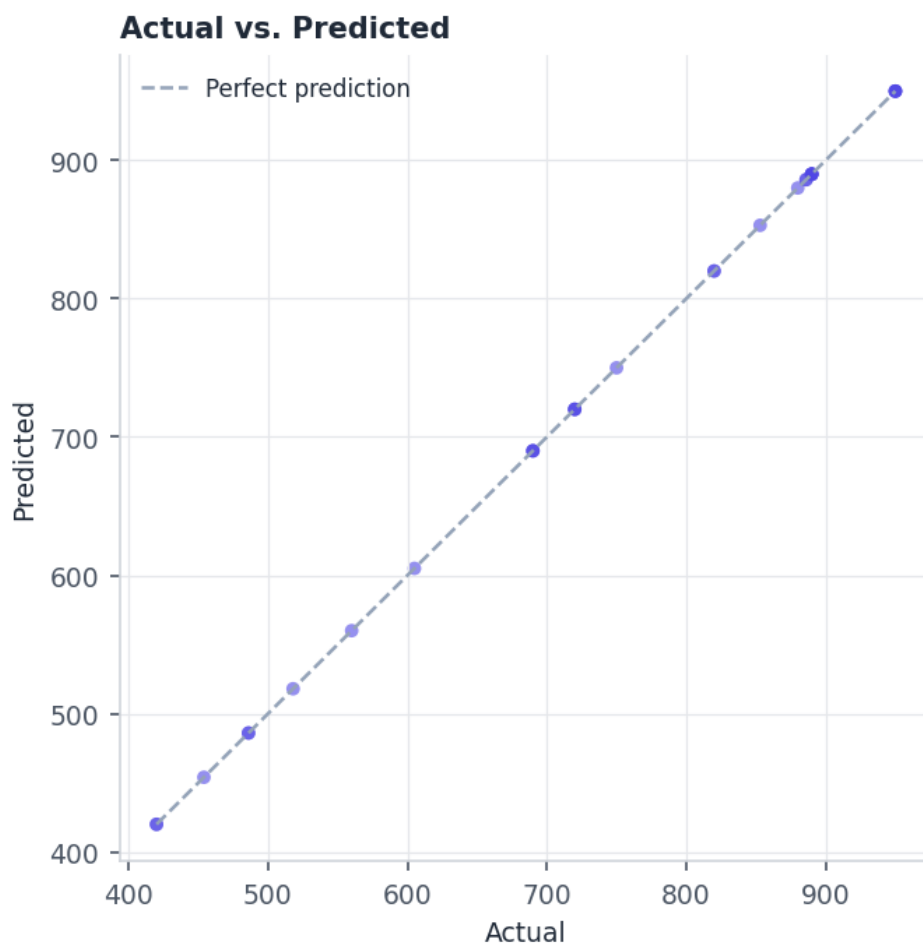


In plain language

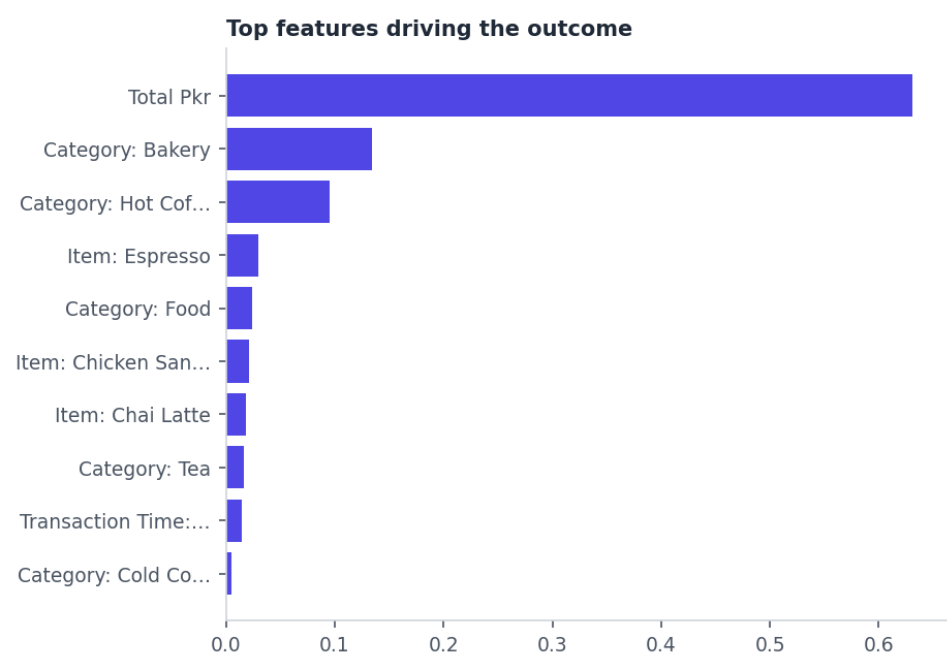
- On average, predictions are off by about 0.00 units (Mean Absolute Error).
- The model explains about 100.0% of the variation in the target ($R^2 = 1.000$).

Compared against a trivial baseline (R^2 (variance explained) = 0.0): this model clearly outperforms it.

Actual vs. Predicted



7. What Drives the Outcome



Feature	Relative importance
Total Pkr	63.1%
Category: Bakery	13.4%
Category: Hot Coffee	9.5%
Item: Espresso	3.0%
Category: Food	2.4%
Item: Chicken Sandwich	2.1%
Item: Chai Latte	1.8%
Category: Tea	1.6%
Transaction Time: Year	1.4%
Category: Cold Coffee	0.5%

8. Future Outlook

- 'Total Pkr' is the strongest single predictor of Unit Price Pkr in this model (relative importance 63.1%).
- 'Category: Bakery' is the second most influential predictor of Unit Price Pkr in this model (relative importance 13.4%).
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Example verified predictions

Actual	Predicted
890.00	890.00
890.00	890.00
950.00	950.00
720.00	720.00
690.00	690.00

9. AI Business Insights AI-generated

AI insights are unavailable right now (gemini: Gemini rate limit exceeded — too many requests. Try again shortly.; ollama fallback: The AI service is currently unavailable.). The rest of this report contains verified analytical results only.

10. Limitations

- Model predictions are estimates based on historical patterns and may not hold if underlying business conditions change. They should inform, not replace, domain expertise.

11. Appendix

Full model metrics

Metric	Value
Mean Absolute Error (MAE)	0.0007
Mean Squared Error (MSE)	0.0052
Root Mean Squared Error (RMSE)	0.0718
R ² (variance explained)	1.0
Mean Absolute Percentage Error	0.0
Cv Mean	1.0
Is Weak	False

Preprocessing

Setting	Value
Test Size	0.2
Cv Folds	5
Random seed	42
Model version	1

Hyperparameters

Parameter	Value
Random Seed	42

Dataset statistics

Statistic	Value
Rows	69,785
Columns	12